

Optimal Line-Drawing for Legal Rules

Jacob Goldin* David Weisbach[†]

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Abstract

Line drawing is a pervasive feature of legal systems. Across many domains, the applicable legal treatment turns on whether an activity falls above or below some threshold. We characterize the economic forces that determine the optimal placement of such thresholds, both generally and in specific legal settings involving monetary and non-monetary sanctions, prohibitions, and mandates. We apply our results to two important line-drawing questions in private law: the optimal due-care threshold for negligence liability and the optimal threshold for whether a shortfall in contract performance constitutes breach. In both applications, the optimal line takes the form of a population-level marginal cost-benefit rule, with the relevant costs and benefits weighted over the actors whose behavior the line induces to change.

Keywords line drawing, legal thresholds, negligence, contract breach

*University of Chicago and NBER.

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1 Introduction

Line drawing is a pervasive feature of legal design. Across many domains, the applicable legal treatment turns on whether some person, product, activity, or institution falls above or below a threshold. For example, environmental law subjects firms to different requirements once emissions or output cross specified levels. Labor and employment law rules turn on employee counts, hours worked, or earnings. Licensing, zoning, health-and-safety regulation, financial regulation, tax law, and safety net programs likewise rely on categorical boundaries that trigger different legal consequences. In private law, negligence liability depends on whether a defendant's behavior crosses a due-care threshold. Similarly, the availability of contract damages hinges on the question of whether a party's contractual performance was sufficiently lacking to constitute a breach.

Whenever legal treatment changes at a threshold, both the location of the line and the legal consequence of crossing it shape actors' behavior. The nature of the legal consequences determine how strongly actors seek to remain on the favorable side of the line. The line's location determines which actors adjust their behavior and by how much. Optimal line-drawing therefore requires balancing the benefit of deterring socially harmful conduct against the cost of distorting private behavior.

This paper develops an economic framework for analyzing optimal line drawing. We study settings in which actors differ in their privately preferred level of some activity, and in which the activity also generates social costs or benefits that actors do not fully internalize. The government influences behavior by treating conduct on one side of the line differently from conduct on the other. Within this setting, we ask where the line should be drawn and how the legal consequences attached to crossing it should be designed. Rather than analyzing each legal regime in isolation, we develop a single framework that nests monetary sanctions, non-monetary sanctions, mandates, and prohibitions. We show that optimal line drawing across these otherwise disparate settings is governed by a common set of tradeoffs.

The central idea is that lines cause actors to sort into three groups: those whose preferred conduct already falls on the favorable side of the line and are unaffected by

it; those who alter their conduct just enough to avoid crossing, thereby “bunching” at the line; and those who choose a level of conduct on the unfavorable side of the line, bearing the legal consequences that apply there. Moving the line or changing the stakes attached to crossing it can change who bunches as well as the costs and benefits of bunching.

We first show that, for any given set of legal consequences attached to crossing the line, moving the threshold affects welfare through two distinct groups of actors. First, moving the line changes the behavior of actors who continue to bunch at it, altering both the private burden they bear and the external harm their conduct causes. Second, moving the line changes whether the marginal actor bunches or instead crosses the line. For example, if the line is a speed limit, raising the limit benefits drivers who already slow down to comply by reducing how much they must slow down, but increases the accident risk they impose. At the same time, a slightly higher limit makes it less costly for the marginal speeder to slow down, inducing that driver to do so. The optimal speed limit therefore balances the effects on drivers who already slow down against what society gains or loses when one more driver chooses to slow to the limit rather than continue speeding.

We then characterize the optimal private stakes for crossing the line. Suppose the government can add a fixed monetary fine or subsidy to whatever other legal consequences apply. Because this fixed amount changes whether the marginal actor bunches or crosses the line without changing the conduct of actors who continue beyond it, it can be used to align the marginal actor’s private choice with the socially preferred outcome. At the optimum, society is indifferent between having that actor bunch and having the actor cross the line. Once the stakes are chosen to satisfy this condition, the marginal-switching effect drops out of the condition for the optimal line, so that the line’s optimal placement depends only on its effects on actors whose behavior it continues to change.

This result has an important implication for generally applicable legal rules. The optimal line is not determined by the costs and benefits for the regulated population as a whole. Instead, the stakes for crossing the line determine which actors are induced to alter their behavior in order to remain on the favorable side of it, and

the location of the line then balances the costs and benefits of changing behavior among those actors. Thus, although a legal threshold may formally apply to an entire population, its optimal placement is governed by a narrower group: the actors whose conduct the line actually changes.

We next apply these results to several common kinds of legal rules. With a fixed monetary sanction, the optimal line equates the marginal external harm at the threshold with the average marginal private burden among the actors induced to bunch there. A partial prohibition – i.e., a rule that prohibits conduct beyond some threshold – produces the same optimal line-drawing condition because everyone whose preferred conduct lies above the line is constrained by it. Nonmonetary sanctions differ because they consume real resources rather than merely transferring money; increasing the sanction may induce additional bunching, but it also increases the social cost imposed on those who continue to cross the line. Mandates require actors who cross the line to take some precaution or adopt some technology. Their design therefore depends not only on the cost of compliance but also on how effectively the mandate reduces the harm generated beyond the line.

To illustrate the framework in detail, we apply it to two private-law settings: negligence and contract. In negligence law, a due-care standard draws a line between negligent and nonnegligent conduct, while damages determine the stakes attached to falling short of that standard. Because damages are paid only when an accident occurs, a risk-neutral actor's behavior depends on expected liability. The optimal uniform due-care standard is a population analogue of the Hand Rule: it balances the additional accident risk created by relaxing the standard against the average burden of precaution among the actors whose conduct the negligence line induces to change. We then show that the same population-level logic applies to contractual performance when a common performance standard cannot be conditioned on the performing party's privately observed costs. The optimal performance requirement balances the marginal value of better performance against the average marginal burden among the parties whose performance the contractual standard induces to change.

Although line drawing is a recurring theme in legal scholarship, most prior work has focused on particular legal domains rather than on identifying the general consid-

erations that should govern the design of legal thresholds. A wide range of scholarship has analyzed what are, in effect, line-drawing problems in areas as diverse as constitutional law, intellectual property, and remedies (Schauer, 1991). This work typically focuses on the contours of particular rules or doctrines and evaluates those lines by reference to domain-specific objectives, such as fidelity to formal legal definitions or consistency with the perceived purpose of the underlying law.

Economic analyses of line drawing have likewise tended to be tied to specific institutional settings. One prominent example is the choice of the due-care standard in negligence law, which can be understood as a form of line drawing; see Shavell (2009). A related literature in public finance studies the optimal firm size threshold to be exempt from value-added tax registration (Keen and Mintz, 2004; Kanbur and Keen, 2014; Brashares et al., 2014) or other tax administration requirements (Dharmapala et al., 2011). Also in tax law, and closer to our focus, Weisbach (1999), Weisbach (2000), and Gillitzer et al. (2017) study line drawing through the lens of minimizing distortions among otherwise similar activities subject to differential tax treatment. Our paper builds on these literatures by analyzing line drawing within a general economic framework in which the regulated activity may generate either social harms or social benefits, so that differentiated legal treatment can be justified on welfare grounds.

Closely related to our formal analysis is Kaplow (2019), who studies the optimal setting of size-based exemptions from regulation. As in our setting, actors make privately optimal decisions that generate externalities, may alter their behavior to remain on the favorable side of a regulatory threshold, and the government chooses the threshold to maximize social welfare. Our analysis differs from Kaplow’s primarily in its focus on the broader class of legal consequences that may attach to crossing a line. Whereas Kaplow studies exemptions from regulations intended to reduce firms’ externality production, our framework encompasses monetary and nonmonetary sanctions, mandates, and partial prohibitions. This broader framework allows us to identify the common structure underlying optimal line drawing across these different legal settings and, in particular, to distinguish the role of the line itself from the role of the stakes attached to crossing it.

A parallel literature in law and economics considers other features of the boundary between legal regimes, especially the choice between rules versus standards (e.g., Kennedy, 1975; Schlag, 1985; Schauer, 1991; Sunstein, 1995). Kaplow (1992) provides the standard framework for this choice, arguing that the primary difference between rules and standards is whether costs are incurred *ex ante* or *ex post*. More recent contributions show how standards can induce beneficial selection (Friedman and Wickelgren, 2014) or mitigate distortions from sharp discontinuities (Bubb and Dari-Mattiacci, 2025). To our knowledge, none of the papers in this large literature address the optimal location for where to draw the boundary itself – whether that boundary takes the form of a rule or a standard – our focus here.

A related question of legal design concerns *how many* lines to draw, or in the limit, whether to rely on a sharp line threshold versus a more gradual, smoother approach to differentiating between legal regimes (e.g., Alexander, 2008; Kolber, 2014; Fennell, 2019).¹ Prior work addresses aspects of this issue in the abstract as well as in specific legal contexts such as criminal law (Kolber, 2015) and tax policy (Slemrod, 2013). As with the rules-versus-standards literature, these analyses typically take as given both the location of thresholds and the consequences associated with crossing them. Our contribution is complementary: we study how the location of the threshold should be chosen jointly with the legal consequences attached to crossing it, and how those conditions vary across different forms of legal treatment.

Finally, a large empirical literature in public economics studies behavioral responses to discontinuities in policy, often using bunching at thresholds to estimate structural preference parameters; see Kleven and Waseem (2013) for a review. Closer to our focus, Sallee and Slemrod (2012) quantify the welfare losses from fuel economy regulations that rely on multiple discrete thresholds relative to smoother policy instruments, and Gillitzer et al. (2017) study how firms design products to fall on the favorable side of tax discontinuities. These papers differ from us in that, while they study the behavioral effects of legal thresholds, they do not provide results relating

¹A distinct question concerns the institutional mechanism through which lines are implemented, such as rule-making versus adjudication. For an analysis in the tax context, see Givati (2014).

to the optimal design of the threshold itself.²

The remainder of the paper proceeds as follows. Section 2 introduces the line-drawing problem informally, surveys the principal forms of differentiated legal regimes, and develops the basic intuition using a speed limit as a running example. Sections 3 and 4 then develop the general framework. Section 3 sets out the model and characterizes how agents sort among remaining unaffected, bunching at the line, and choosing the strict regime. Section 4 derives the general conditions governing the optimal threshold and the optimal stakes for crossing it. Section 5 specializes these results in turn to fixed monetary sanctions, partial prohibitions, nonmonetary sanctions, and regulatory mandates. We then apply the framework to two private-law settings. Section 6.1 derives the jointly optimal due-care threshold and damages schedule under negligence liability, and Section 6.2 develops the corresponding results for contractual performance thresholds and expectation damages. Section 7 concludes.

2 Motivation and Overview

Before turning to our formal analysis, this section informally describes the class of line-drawing problems on which we will focus as well as the intuition for our main results.

2.1 The Line-Drawing Problem

We focus on line-drawing problems characterized by four main features. First, agents make some choice about how to behave and that choice generates externalities. Familiar examples include: a driver who chooses how fast to drive imposes accident risk on others; a firm that chooses a level of production generates pollution; a party to a contract chooses a level of performance that affects its counterparty; a manufacturer

²Although it does not consider the optimal location for a threshold, Bergstrom et al. (2025) is closer to our focus; it studies how behavioral responses to a reform in the location of a notch can help identify the reform’s welfare effect.

chooses how much to invest in product safety. In each case, absent a legal intervention, an agent left to its own devices weighs only its private costs and benefits, so behavior that is harmful to others tends to be supplied in excess of the socially desirable level.

The second defining feature of our setting is that the law responds to the externality-generating behavior through a *differentiated legal regime*. By this, we mean that the law sorts actors into a legal category based on their chosen activity level and imposes a corresponding legal consequence based on the assigned category. We refer to the more favorable (to agents) category as the *lax regime* and the less favorable category the *strict regime*. Under a negligence regime, for instance, an actor whose activity meets the standard of due care falls into the lax regime and bears no liability, while an actor who fails to exercise due care falls into the strict regime and is liable for damages. The legal consequence of falling into the strict regime is some obligation or treatment the actor would prefer to avoid: to pay money, to bear a nonmonetary cost such as lost time or liberty, or to undertake a required action such as adopting a technology or making a disclosure. The private burden generated by these legal consequences determines the stakes associated with crossing the line. The consequence may be *fixed*, in that it does not change with the actor's conduct once the threshold is crossed, or *variable*, in that it varies based on where the actor's conduct falls within the strict regime. As we will see, both the form of the consequence and whether it is fixed or variable shape the optimal line-drawing analysis.

The third feature of our setting is that the rule that assigns actors to legal regimes is a discrete threshold defined with respect to the externality-generating behavior. This threshold is the *line*. A defining feature of the line is its all-or-nothing nature: an actor whose behavior falls to one side of the line is governed by the lax regime whereas an actor whose behavior is on the other side of the line is governed by the strict regime.

Finally, we focus on lines that are *generally applicable*. The threshold assigns actors to legal regimes based on their chosen behavior rather than an individualized assessment of the particular costs and benefits associated with that behavior for each actor. This restriction may arise because the characteristics relevant to such an indi-

vidualized determination are unobservable to the government or because accounting for them would be too costly or burdensome to administer. As a result, the same line applies to actors who may differ substantially in their private circumstances.

2.2 Types of Legal Regimes

Across differentiated legal regimes, there is wide variation in the nature of the legal consequences imposed as a result of crossing the line.

Monetary sanctions. A monetary sanction, such as a fine or penalty, entails a transfer of resources from the actor to others in society. Setting aside potential distributional concerns that could shape the desirability of such transfers, the imposition of a monetary sanction does not constitute a direct social welfare cost or benefit.

Monetary sanctions may be fixed or variable. A parking ticket is the paradigm of a fixed monetary sanction. In Chicago, the fine for failing to feed the meter is currently \$50 (\$70 in the central business district) no matter how long the car remains in the spot, at least up to a broad limit. The fine for running a stop sign in Illinois is \$164 regardless of how close the driver comes to stopping. An officer-issued ticket for speeding is \$164 for any speed up to 25 miles per hour over the limit, so that the speed limit functions as a fixed fine for all but the most extreme speeding. Some contract damages fit the same pattern. For example, stipulated damages may impose a fixed obligation once a contracting party's performance deteriorates enough to constitute a breach.

The distinctive feature of a fixed monetary sanction is that conditional on crossing the line, the fine does not change. As a result, actors who cross the line have no reason to moderate their conduct further. Therefore, fixed monetary sanctions create an incentive to bunch at the line but actors who stay above the line will choose the same behavior that they would absent the sanction.

In contrast, variable monetary sanctions scale based on the activity level within the strict regime. With negligence liability, for example, actors whose conduct falls on the negligent side of the due-care threshold owes damages that rise with the harm

caused. Contracts for which damages track the extent of breach fit here as well. Many statutory fines are also variable; for example, OSHA penalties vary according to an elaborate schedule measuring the severity of the violation.

The key economic difference from the fixed monetary sanction case is that actors who choose to remain above the line now have a reason to alter their conduct. Altering conduct such as by taking more care or reducing activity levels reduces the sanction the actor bears. As a result, variable sanctions operate along two margins: which actors bunch at the line and how they affect the behavior of actors who remain above it.

Nonmonetary sanctions. The defining feature of a nonmonetary sanction is that it imposes a cost on the actor in some form other than money. Imprisonment is the classic case, with the definition of the offense serving as the line and the prison term the sanction. Hassle is also a nonmonetary sanction. For example, making a speeder show up in court rather than merely paying a fine is a form of nonmonetary sanction. Less conventional sanctions would also qualify, such as “shame lists” that publicly disclose the identities of heavily delinquent taxpayers in a jurisdiction.

As with monetary sanctions, nonmonetary sanctions can be fixed or variable. Unlike monetary sanctions, nonmonetary sanctions consume real resources – they are not simply transfers from one party to another. The optimal policy analysis therefore recognizes the imposition of a nonmonetary sanction as a direct social welfare cost in determining a line’s optimal placement.

Mandates. The third category of differentiated legal regime we consider is a requirement that certain actors undertake a particular action, such as to adopt a specified technology or procedure. For example, certain polluters, but not others, may be required to install abatement equipment. Similarly, banks above a certain size may be subjected to more stringent capital requirements. In these settings, an actor is subjected to the requirement based on whether their conduct is above the applicable line, and the legal consequence of crossing the line is the obligation to engage in the required action.

Like the other regimes we consider, mandates can be fixed or variable, depending on the nature of the obligation imposed on actors in the strict regime. Fixed mandates might include requirements to install a railing, an emergency exit, or an ADA-compliant ramp; or to disclose certain pieces of information such as a large campaign contribution or acquisition of a large equity stake in a publicly traded company. Other mandates scale with activity; for example, a bank's required capital cushion might increase with its size, or the cost of a firm's required pollution abatement technology could rise with the level of the firm's output.

Like a nonmonetary sanction, a mandate imposes a real resource cost on actors subjected to the strict regime. Unlike a monetary sanction, the cost associated with a mandate is generally intended to reduce the harm from remaining above the line. For example, abatement equipment lowers pollution. Capital requirements lower the risk of a bank failure.

Partial Prohibitions, Property Rules, and Injunctions. The final category of differentiated legal regime we consider is what we call a partial prohibition. We use this term to refer to a rule under which conduct in the lax regime remains available but conduct that would place an actor in the strict regime is prohibited. Thus, unlike a sanction or mandate, a partial prohibition does not allow an actor to cross the line and simply bear the resulting legal consequence. The prohibition is “partial” because it bars only conduct beyond the line rather than the underlying activity altogether. Licensing requirements are one example: an actor who meets a minimum competency threshold may sell a service (the lax regime), while one who does not is barred from selling it (the strict regime). Product-safety rules have the same shape, permitting any product at or above a minimum safety level and prohibiting those below it. Other examples include building-height limits, strict emissions limits on polluting devices, occupational-exposure limits; the minimum-wage and minimum-workplace-safety rules.

The key feature of partial prohibitions is that no actors can be in the strict regime. Everyone is either below the line or bunched at the line. For this reason, a partial prohibition can be understood as the limiting case of either a monetary or

nonmonetary sanction regime where the sanction is set high enough that no actor chooses to cross the line.

2.3 Intuitions for Optimal Line Drawing

In all of the settings we consider, lines sort actors into three groups: (i) those who prefer to be below the line; (ii) those who, all else equal, prefer to be above the line but alter their behavior to stay below the line; and (iii) those who prefer to be above the line and choose to remain above the line. To illustrate with a simplified example, consider a speed limit. Some drivers prefer to travel at or below the limit and would do so even if no limit existed; the line does not affect them. A second group would prefer to drive faster but slows down because the expected cost of a ticket exceeds the benefit of arriving sooner at their destination. Because these drivers would prefer to drive faster, they reduce their speed only as much as necessary to comply with the speed limit. We say that these drivers *bunch* at the line. The *marginal buncher* is the driver who is just indifferent between slowing to the limit and accepting the sanction. The third group prefers the risk of paying the fine to slowing down and continues to speed. When the sanction for violating the speed limit is fixed, this last group has no reason to moderate its behavior: once a driver has decided to be in the strict regime (i.e., to speed), the driver may as well travel at their privately preferred speed.³

Dividing actors into these three groups helps guide the welfare analysis. Consider a small change to the line (e.g., a slight increase or decrease in the speed limit). Moving the line has two distinct effects. First, it changes the behavior of actors who remain in the bunching group: when the line moves, these actors move with it. We refer to this as the *inframarginal-buncher effect*. Second, moving the line changes whether the marginal buncher bunches or instead remains in the strict regime. Because these two choices can involve substantially different behavior and legal consequences, moving the marginal actor between them can produce a discrete

³In practice, the probability of detection or enforcement may rise with speed, so the expected sanction may be variable rather than fixed.

change in social welfare. We refer to this as the *upper-edge effect*.

The simplest case, and our benchmark, is a fixed, monetary sanction, such as the speed limit. Drivers who prefer to stay below the speed limit are unaffected. The bunched drivers reduce their speed to the limit. Slowing down reduces the accident risk they impose on others but also makes the bunched drivers worse off because they arrive at their destinations later. Those who continue to speed pay the fine, but the payment is a transfer and therefore does not itself reduce social welfare. Moreover, because these drivers do not slow down, they continue to impose the same external harm. In this fixed-monetary-sanction benchmark, therefore, all of the behavioral and welfare effects occur among the bunched drivers.

The optimal fine for this case has a simple interpretation. Raising the fine expands the number of bunched drivers. At the optimal policy, the fine selects a marginal driver for whom the private loss from slowing to the limit just equals the reduction in external harm produced by doing so. Because that marginal driver is privately indifferent about whether to bunch, the magnitude of the optimal fine equals the driver's private loss from slowing.

Given the optimal fine, the optimal location of the line balances two forces. On the one hand, raising the speed limit allows all bunched drivers to drive closer to their preferred speeds, improving their private welfare. At the same time, raising the speed limit permits more of the harmful behavior (faster driving) the law is intended to curb. The optimal line therefore equates the marginal increase in accident harm from raising the limit with the average marginal reduction in private utility among the drivers who are induced by the law to lower their driving speeds.

When the strict regime imposes real costs or changes the behavior of those who remain above the line, the upper-edge effect can become important. Consider, for example, a negligence regime. An actor who remains negligent may nevertheless take additional care because doing so reduces expected liability for damages. Similarly, a polluter subject to a regulatory mandate may alter its emissions or production decisions in response to the costs created by the mandate. Moving the marginal actor between bunching and the strict regime therefore changes not only that actor's activity but also any costs or benefits associated with the strict regime. A nonmone-

tary sanction or compliance obligation may consume real resources, making bunching relatively more desirable. Conversely, a mandate may reduce external harm, making remaining in the strict regime relatively more desirable.

An interesting result is that the upper-edge effect disappears if the government can also choose a fixed monetary fine or subsidy attached to the strict regime. As discussed above, a monetary payment changes actors' incentives to bunch without changing behavior conditional on crossing the line and, because it is a transfer, without directly affecting social welfare. When chosen optimally, the payment makes the marginal actor socially indifferent between bunching and remaining in the strict regime. Moving that actor from one regime to the other then has no first-order welfare effect. The optimal line is therefore determined entirely by its effects on actors who continue to bunch.

3 General Framework

This section describes our formal model.

3.1 Environment

There is a population of agents of mass 1. Each agent chooses an action $x \in [0, 1]$.

The government regulates agents' actions through a differentiated legal rule in which there are two regimes: a *lax regime*, denoted by L , and a *strict regime*, denoted by S . We focus on the threshold $t \in [0, 1]$, which is the line that separates the two legal regimes. Actions below the line, $x \leq t$, are governed under the lax regime, whereas actions above the line, $x > t$, are governed under the strict regime.⁴

Let $v(x; x^*)$ denote the agent's private payoff from selecting action x under the lax regime. We interpret x^* as the agent's privately optimal action under the lax

⁴We normalize the action so that the strict regime applies above the threshold. This is without loss of generality: a rule that applies to values below a threshold can be represented by reversing the action scale, for example by defining x as a shortfall from care rather than as care itself.

regime:

$$x^* = \arg \max_{x \in [0,1]} v(x; x^*). \quad (1)$$

We assume $v(x; x^*)$ is continuously differentiable and strictly concave in x , so that x^* is well-defined. Types (x^*) are distributed according to CDF F with differentiable density f . We assume the government does not observe an agent's type so cannot condition policy upon it. Finally, we assume

$$v_{xx^*}(x; x^*) > 0, \quad (2)$$

which guarantees that higher types (i.e., higher values of x^*) place a greater marginal value on increasing x .

Let $c(x)$ denote the net private burden generated by the legal consequences of choosing an action $x > t$ and thereby being governed under the strict regime. Thus, $c(x)$ captures the private cost of legal consequences such as a monetary payment, a nonmonetary sanction, or compliance with a regulatory mandate. We assume that $c(x)$ is continuous and weakly convex. Depending on the application, this private burden may be fixed or may vary with the chosen action. Because the agent's type x^* is private and not observed by the government, $c(x)$ does not vary directly with type, except insofar as type affects the agent's choice of x . We restrict our focus to settings in which, all else equal, agents prefer to be governed under the lax regime:

$$c(x) > 0 \text{ for all } x. \quad (3)$$

By construction, utility in the strict regime, $u_S(x; x^*)$, is equal to

$$u_S(x; x^*) = v(x; x^*) - c(x).$$

We denote the agent's privately optimal action under the strict regime by:

$$x_S(x^*) = \arg \max_{x \in [t,1]} u_S(x; x^*). \quad (4)$$

Note that neither regime's payoff depends directly on the threshold t ; the threshold determines only which regime applies.⁵

3.2 Behavior and Sorting

To characterize behavior, consider first an agent with $x^* \leq t$. For such an agent, selecting $x = x^*$ is privately optimal, since it results in the agent being governed under the (more desirable) lax regime, and the agent prefers x^* to any other possible choice within that regime.

Next, consider an agent with $x^* > t$. If the agent is governed under the strict regime, its privately optimal choice is to select $x_S(x^*)$. Alternatively, the agent could select some $x \leq t < x^*$ and thereby be in the lax regime. Among this latter category of choices, concavity of v guarantees that bunching at the threshold, i.e., selecting $x = t$ (the lax regime option that is closest to x^*), is privately optimal.

To decide whether to bunch, the agent compares the utility from its privately optimal strict-regime action, $u_S(x_S(x^*); x^*)$, to the utility it would obtain from bunching, $v(t; x^*)$. Let $\Delta_B(t, x^*)$ denote the net utility benefit from bunching,

$$\Delta_B(t, x^*) := v(t; x^*) - u_S(x_S(x^*); x^*).$$

The structure imposed above implies that the benefit from bunching is strictly decreasing in the agent's type. To see this, note that applying the envelope theorem to the agent's optimized utility under the strict regime gives

$$\frac{\partial \Delta_B(t, x^*)}{\partial x^*} = v_{x^*}(t; x^*) - v_{x^*}(x_S(x^*); x^*).$$

For a type whose preferred strict-regime action lies above the threshold, $x_S(x^*) > t$,

⁵One can imagine settings in which this assumption is violated. For example, if agents view t as demarcating the boundary of ethical behavior, those who cross the line may care about the magnitude of their deviation from t even apart from the associated legal consequences. Similarly, agents' ideal points could depend in part on others' behavior; for example, if t is a speed limit that induces bunching, t 's location could affect the speed that non-speeders choose to drive in furtherance of their own safety goals.

this expression can be written as

$$\frac{\partial \Delta_B(t, x^*)}{\partial x^*} = - \int_t^{x_S(x^*)} v_{xx^*}(s; x^*) ds < 0,$$

where the inequality follows from (2). Intuitively, choosing the strict regime permits the agent to select a higher action than bunching at the threshold. Because higher types place a greater relative value on higher actions, they are less willing to bunch.

It follows that the set of bunching types is an interval. We denote the marginal type by $\tilde{x}(t) \in [t, 1]$, defined by

$$\Delta_B(t, \tilde{x}(t)) = 0.$$

The marginal type is indifferent between bunching at the threshold and being in the strict regime. Types with $t < x^* < \tilde{x}(t)$ strictly prefer to bunch at the threshold, while types with $x^* > \tilde{x}(t)$ strictly prefer to be in the strict regime. We treat the marginal type as bunching. If $\Delta_B(t, 1) > 0$, then every type above the threshold prefers to bunch, in which case we set $\tilde{x}(t) = 1$.

Summarizing our discussion so far, agents' behavior can be characterized as follows:

$$x(x^*; t) = \begin{cases} x^* & \text{if } x^* \leq t, \\ t & \text{if } t < x^* \leq \tilde{x}(t), \\ x_S(x^*) & \text{if } x^* > \tilde{x}(t). \end{cases} \quad (5)$$

3.3 Social Welfare

Social welfare may diverge from private utility because an action generates costs or benefits that the agent does not internalize. For example, an action may generate an external cost, $\phi(x)$, and the regulator may draw a line and impose legal consequences on agents in the strict regime in order to reduce or internalize that cost. In addition, a legal consequence may have different effects on private utility and social welfare. The private burden it imposes is captured by $c(x)$, but that private burden need not

equal its social cost. A monetary payment, for example, may be a pure transfer, so that it imposes a private burden without directly lowering social welfare, whereas a nonmonetary sanction or compliance obligation may consume real resources and therefore reduce social welfare as well as private utility.

We represent social welfare directly under each of the three possible behavioral outcomes. Let $W_L(x^*)$ denote the social welfare generated by an agent with type x^* who remains in the lax regime and selects its privately optimal action x^* . Let $W_B(x^*; t)$ denote the social welfare generated by that agent if it instead bunches at the threshold t , and let $W_S(x^*)$ denote the social welfare generated if the agent is in the strict regime and selects its preferred strict-regime action $x_S(x^*)$.⁶ We assume that $W_L(x^*)$ and $W_S(x^*)$ do not depend directly on the threshold and that $W_B(x^*; t)$ is differentiable in t .

These welfare functions incorporate all private and external effects associated with the agent's action and applicable regime. Thus, they include the agent's private payoff from the chosen action, and hence any loss from departing from x^* , as well as any externalities. W_S also incorporates the net social effect of the legal consequences associated with the strict regime, which may differ from the private burden those consequences impose, $c(x)$. In our negligence application, for example, we consider the possibility that being in the strict regime (i.e., being held liable for negligence) requires society to incur a real resource cost from calculating damages.⁷

Using (5), aggregate social welfare induced by a threshold t is therefore

$$W(t) = \int_0^t W_L(x^*) dF(x^*) + \int_t^{\tilde{x}(t)} W_B(x^*; t) dF(x^*) + \int_{\tilde{x}(t)}^1 W_S(x^*) dF(x^*). \quad (6)$$

⁶Equivalently, one could begin with welfare functions $w_L(x; x^*)$ and $w_S(x; x^*)$ defined over actions and obtain W_L , W_B , and W_S by evaluating them at x^* , t , and $x_S(x^*)$, respectively. We suppress this additional layer of notation because only the resulting realized welfare functions enter the analysis.

⁷A different form of administrative cost that our baseline analysis abstracts from is the possibility that it is costly for society to classify agents whose conduct falls near the threshold. We consider this possibility in the Online Appendix and show that our results extend in a straightforward manner.

4 Optimal Policy

This section uses our model to characterize optimal line-drawing and the optimal private stakes for crossing the line.

4.1 Optimal Threshold

Differentiating (6) with respect to t yields

$$\frac{\partial W}{\partial t} = \int_t^{\tilde{x}} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) + \frac{\partial \tilde{x}}{\partial t} [W_B(\tilde{x}; t) - W_S(\tilde{x})] f(\tilde{x}). \quad (7)$$

Setting (7) equal to zero yields the following Proposition.

Proposition 1 (General reduced-form threshold condition). *Any interior optimal threshold satisfies*

$$\int_t^{\tilde{x}} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) = \frac{\partial \tilde{x}}{\partial t} [W_S(\tilde{x}) - W_B(\tilde{x}; t)] f(\tilde{x})$$

Proposition 1 highlights the optimal threshold logic that recurs throughout the paper. The left-hand side is the welfare contribution from changing the behavior of inframarginal bunched. The right-hand side captures what we refer to as the upper-edge effect: $W_S(\tilde{x}) - W_B(\tilde{x}; t)$. It is the welfare difference between the marginal type remaining in the strict regime versus bunching at the threshold. The applications below differ in this upper-edge effect because the legal consequences associated with different strict regimes have different effects on social welfare.

4.2 Optimal Private Stakes for Crossing the Line

Proposition 1 characterizes the optimal location of the threshold without imposing additional restrictions on the legal consequences associated with the strict regime. There is no directly analogous general condition for the optimal legal consequence because legal consequences can play different roles depending on the applicable legal

regime. Changing them may affect whether agents choose to be in the strict regime, the actions they choose in the strict regime, and the social costs or benefits they generate. The optimality condition therefore depends on which aspect of the legal consequences the government is able to adjust.

We focus on a common and useful policy margin. In many settings, the government can impose a fixed monetary fine or subsidy on top of whatever other legal consequences apply in the strict regime. For example, a regulatory mandate or non-monetary sanction may determine the underlying legal treatment of an agent who crosses the line, while a separate fine or subsidy changes the private stakes associated with crossing it without altering those underlying obligations.

Formally, suppose we can write the net private burden associated with the strict regime as

$$c(x; \tau) = c_0(x) + \tau,$$

where τ is a fixed monetary fine or subsidy imposed on agents in the strict regime and $c_0(x)$ reflects the private burden generated by the other legal consequences associated with that regime.⁸ Because τ is a pure transfer, it affects private incentives but does not directly affect social welfare other than through its effect on agents' behavior.

Private utility in the strict regime is

$$u_S(x; x^*, \tau) = v(x; x^*) - c_0(x) - \tau.$$

Because τ does not vary with the action, it does not affect the agent's preferred action conditional on being in the strict regime. Thus,

$$x_S(x^*) = \arg \max_{x \in [t, 1]} \{v(x; x^*) - c_0(x)\},$$

which is independent of τ . In contrast, τ does affect the relative attractiveness of bunching at the threshold or being in the strict regime. In particular, the net private

⁸We restrict attention to values of τ that satisfy (3), i.e., $c_0(x) + \tau > 0$ for all x .

benefit from bunching is

$$\Delta_B(t, x^*; \tau) = v(t; x^*) - v(x_S(x^*); x^*) + c_0(x_S(x^*)) + \tau.$$

An increase in τ therefore makes bunching more attractive for every type without changing behavior for those who remain in the strict regime. Let $\tilde{x}(t, \tau)$ denote the marginal type who is indifferent between bunching and being in the strict regime. For a given threshold t , aggregate social welfare can be written as

$$W(t, \tau) = \int_0^t W_L(x^*) dF(x^*) + \int_t^{\tilde{x}(t, \tau)} W_B(x^*; t) dF(x^*) + \int_{\tilde{x}(t, \tau)}^1 W_S(x^*) dF(x^*). \quad (8)$$

Differentiating (8) with respect to τ yields

$$\frac{\partial W(t, \tau)}{\partial \tau} = [W_B(\tilde{x}; t) - W_S(\tilde{x})] f(\tilde{x}) \frac{\partial \tilde{x}}{\partial \tau}. \quad (9)$$

Setting (9) equal to zero yields the following proposition.

Proposition 2 (General reduced-form stakes condition). *If, in addition to any other legal consequences represented by $c_0(x)$, the government can impose a fixed fine or subsidy τ , then, for a given threshold t , any interior optimal choice of τ satisfies*

$$W_B(\tilde{x}; t) = W_S(\tilde{x})$$

where $\tilde{x} = \tilde{x}(t, \tau)$ denotes the marginal type who is indifferent between bunching and being in the strict regime.

Proposition 2 states that the private stakes for choosing between regimes should be set so that society is indifferent between the marginal type bunching at the threshold and remaining in the strict regime. In other words, τ should be set so that the upper-edge effect is zero.

A useful way to interpret Proposition 2 is through the wedge between private and social incentives. If being in the strict regime generates external harm that agents do not internalize, agents will be too willing to cross the line relative to the social

optimum. In that case, τ can be used to increase the stakes of crossing and thereby expand bunching. Conversely, if the private burden associated with the strict regime arises from a socially beneficial obligation, such as an externality-reducing mandate, agents may avoid it too aggressively, in which case a subsidy may be desirable. More generally, τ adjusts the private stakes associated with crossing the line so as to align private and social comparisons at the margin: it is chosen so that the marginal type—who is privately indifferent between bunching and the strict regime—is also socially indifferent between them.

Combining Proposition 1 and Proposition 2 yields a simplified expression for the optimal threshold. If the government optimally sets the private stakes for crossing the line by choosing τ according to Proposition 2, we obtain:

Corollary 1 (Simplified optimal threshold condition). *If τ is set at an interior optimum according to Proposition 2, then any interior optimal threshold satisfies*

$$\int_t^{\tilde{x}(t)} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) = 0.$$

Corollary 1 shows that if the government can optimally set the private stakes for crossing the line through the fixed monetary component, selecting the optimal threshold simplifies to considering the effect of the threshold on the inframarginal bunchers. Returning to the optimal speed limit problem, for example, this result implies that if the fine for speeding is set optimally, the choice of speed limit should balance the accident prevention benefits versus the private costs of those drivers who choose to slow down because of the speed limit.

5 Application to Specific Differentiated Legal Regimes

This section applies our general results to several specific forms of differentiated legal regimes.

5.1 Fixed Monetary Sanctions

Suppose that crossing the threshold triggers a fixed sanction $\tau \geq 0$ and that action x generates external harm $\phi(x)$. Because the sanction is action-independent, agents who cross the threshold choose their privately preferred action:

$$x_S(x^*) = x^*.$$

Treating the monetary sanction as a transfer, realized welfare is

$$W_B(x^*; t) = v(t; x^*) - \phi(t)$$

for buncers and

$$W_S(x^*) = v(x^*; x^*) - \phi(x^*)$$

for an agent who is in the strict regime.⁹ The marginal bunching agent is privately indifferent between remaining in the strict regime while paying the sanction and bunching at the line:

$$v(\tilde{x}; \tilde{x}) - \tau = v(t; \tilde{x}),$$

or equivalently,

$$\tau = v(\tilde{x}; \tilde{x}) - v(t; \tilde{x}), \tag{10}$$

where $\tilde{x} = \tilde{x}(t, \tau^*(t))$.

Proposition 2 implies that, at an interior optimal choice of τ , the same type is socially indifferent between bunching and choosing the strict regime:

$$\begin{aligned} 0 &= W_B(\tilde{x}; t) - W_S(\tilde{x}) \\ &= v(t; \tilde{x}) - \phi(t) - v(\tilde{x}; \tilde{x}) + \phi(\tilde{x}). \end{aligned} \tag{11}$$

⁹An immediate implication is that in the limiting case in which $\phi(x) = 0$ for all x , but the sanction is fixed at some $\tau > 0$, Proposition 1 implies that the optimal threshold is placed to minimize $\int_t^{\tilde{x}(t)} v(x^*; x^*) - v(t; x^*) dF(x^*)$, where $v(x^*; x^*) - v(t; x^*)$ represents the private harm from bunching. Intuitively, because changing behavior produces no offsetting social benefit, the line should be placed where the classification creates the least aggregate private distortion (see Weisbach, 1999, 2000).

Combining (10) and (11) yields

$$\tau^*(t) = \phi(\tilde{x}) - \phi(t). \quad (12)$$

Thus, the optimal sanction sets the private stakes for crossing the line equal to the social harm avoided when the marginal type chooses to bunch.

To understand the factors that shape the optimal sanction in this setting, consider a second-order Taylor approximation of $\phi(\cdot)$ around t :

$$\tau^*(t) \approx \frac{2[\phi'(t)]^2}{-v_{xx}(t; t)} \quad (13)$$

Higher marginal external harm, ϕ' , raises the value of inducing a behavior change and therefore raises the optimal sanction. Conversely, greater curvature of private utility around the privately preferred action makes any deviation from one's private ideal action more costly and therefore lowers the optimal sanction.

Because the optimal sanction eliminates the upper-edge effect in Proposition 1, Corollary 1 implies that the optimal threshold satisfies

$$\int_t^{\tilde{x}} [v_x(t; x^*) - \phi'(t)] dF(x^*) = 0, \quad (14)$$

or equivalently,

$$\phi'(t) = v_x^B, \quad (15)$$

where v_x^B denotes the average private cost of deviating from one's ideal action among the agents who bunch at the threshold,

$$v_x^B = \frac{\int_t^{\tilde{x}(t)} v_x(t; x^*) dF(x^*)}{F(\tilde{x}(t)) - F(t)}. \quad (16)$$

The optimal threshold formula therefore equates the welfare effect of the marginal change in the externality at the threshold, $\phi'(t)$, to the average private welfare effect caused by shifting behavior among agents who continue to bunch.

5.2 Partial Prohibitions

A partial prohibition can be interpreted as the limiting case of a monetary sanction in which the sanction is so severe that no actor chooses to be in the strict regime. The bunching region therefore extends over all types with $x^* > t$, so that there is no upper-edge effect, i.e., $\frac{\partial \tilde{x}}{\partial t} = 0$. Proposition 1 therefore simplifies to the same expression as in (15). Intuitively, with a partial prohibition, the optimal threshold balances the marginal increase in harm from relaxing the prohibition against the average reduction in distortion for the agents who continue to be constrained by it.

5.3 Nonmonetary Sanctions

Many legal sanctions impose real resource costs rather than simply being monetary transfers. Prison sentences are the canonical example, but waiting costs, administrative burdens, and other forms of costly legal treatment may have the same structure. They often vary with the extent of a violation but can also be fixed.

Suppose that the private burden $c(x)$ imposed by the legal consequence associated with the strict regime constitutes a real resource cost to society. Thus,

$$W_B(x^*; t) = v(t; x^*) - \phi(t)$$

and

$$W_S(x^*) = v(x_S(x^*); x^*) - \phi(x_S(x^*)) - c(x_S(x^*)).$$

By construction, the marginal type remains indifferent between bunching and being governed under the strict regime, so

$$v(t; \tilde{x}) = v(x_S(\tilde{x}); \tilde{x}) - c(x_S(\tilde{x})).$$

Hence, the upper-edge welfare effect is given by

$$W_S(\tilde{x}) - W_B(\tilde{x}; t) = \phi(t) - \phi(x_S(\tilde{x})).$$

Proposition 1 therefore implies that the optimal threshold for a given nonmonetary sanction satisfies

$$\int_t^{\tilde{x}} [v_x(t; x^*) - \phi'(t)] dF(x^*) = \frac{\partial \tilde{x}}{\partial t} [\phi(t) - \phi(x_S(\tilde{x}))] f(\tilde{x}).$$

With a fixed nonmonetary sanction, we have $c(x) = \tau$ and $x_S(x^*) = x^*$. If the government sets the level of this fixed nonmonetary sanction optimally,¹⁰ differentiating welfare with respect to τ and setting the derivative equal to zero yields $\frac{\partial \tilde{x}}{\partial \tau} [\phi(\tilde{x}) - \phi(t)] f(\tilde{x}) = 1 - F(\tilde{x})$. Differentiating the marginal-type indifference condition with respect to t and τ further implies that

$$\frac{\partial \tilde{x} / \partial t}{\partial \tilde{x} / \partial \tau} = v_x(t; \tilde{x}).$$

Combining this relationship with the optimal-sanction condition and the threshold first-order condition yields

$$\int_t^{\tilde{x}} [v_x(t; x^*) - \phi'(t)] dF(x^*) = -v_x(t; \tilde{x}) [1 - F(\tilde{x})],$$

or equivalently,

$$\phi'(t) = v_x^B + v_x(t; \tilde{x}) \frac{1 - F(\tilde{x})}{F(\tilde{x}) - F(t)},$$

where v_x^B is defined as in (16). Thus, unlike with a monetary sanction, setting the nonmonetary sanction to its optimal level does not eliminate the upper-edge effect. Instead, relaxing the threshold has an additional social benefit because some agents who would otherwise remain in the strict regime avoid incurring the real resource cost generated by the sanction. Relative to the corresponding condition for an optimally chosen monetary sanction, this additional consideration therefore pushes toward a

¹⁰Although frequently applied in practice, the optimal nonmonetary sanction can be zero if other policy tools (such as monetary sanctions) are also available (see Polinsky and Shavell, 1984; Shavell, 1987), or sufficiently large that it functions as a partial prohibition (Kaplow, 1990). Our analysis provides a necessary condition for an interior optimal nonmonetary sanction but does not establish that such an interior optimum exists.

higher, more permissive threshold, all else equal, thereby reducing the number of agents who are in the strict regime and incur the socially costly sanction.

5.4 Mandates

Consider next a mandate that requires the adoption of some technology with fixed adoption cost, $c(x) = c$. The mandated technology reduces the external harm generated by one's activity. Denote the reduction in external harm by $\Delta_\phi(x) \geq 0$. Because the cost of the mandate is fixed, it does not affect behavior (conditional on it applying), so

$$x_S(x^*) = x^*.$$

Realized welfare for an agent who remains in the strict regime is therefore

$$W_S(x^*) = v(x^*; x^*) - \phi(x^*) - c + \Delta_\phi(x^*),$$

whereas realized welfare for an agent who bunches remains $W_B(x^*; t) = v(t; x^*) - \phi(t)$. By construction, the marginal type satisfies $v(\tilde{x}; \tilde{x}) - v(t; \tilde{x}) = c$. Substituting these results into Proposition 1 implies that the optimal threshold satisfies

$$\int_t^{\tilde{x}} [v_x(t; x^*) - \phi'(t)] dF(x^*) = [\phi(t) - \phi(\tilde{x}) + \Delta_\phi(\tilde{x})] f(\tilde{x}) \frac{\partial \tilde{x}}{\partial t}. \quad (17)$$

Rearranging (17) and applying (16) allows us to highlight the effect of the harm reduction on the optimal threshold:

$$\phi'(t) = v_x^B + [\phi(\tilde{x}) - \frac{\phi(t)]f(\tilde{x})}{F(\tilde{x}) - F(t)} \frac{\partial \tilde{x}}{\partial t} - \frac{\Delta_\phi(\tilde{x}) f(\tilde{x})}{F(\tilde{x}) - F(t)} \frac{\partial \tilde{x}}{\partial t}$$

Because $\frac{\partial \tilde{x}}{\partial t} \geq 0$, a larger harm reduction from the mandated technology makes leaving the marginal type in the strict regime more socially valuable. Relative to an otherwise identical nonmonetary sanction that produces no harm-reduction benefit, this pushes toward a lower threshold, all else equal, thereby expanding the set of agents subject to the mandate.

The same reasoning extends to mandates for which the cost of complying varies with the action, such as output-dependent pollution-control requirements. The only difference is that when $c(x)$ varies with x , the mandate also affects behavior within the strict regime, so $x_S(\tilde{x})$ replaces $\tilde{x}$ in the upper edge effect.

5.5 Summary

The applications above illustrate how the nature of the legal consequences of being in the strict regime affects the optimal line. With an optimally chosen monetary sanction, the upper-edge effect is eliminated, so the optimal threshold depends only on the welfare effects for agents who continue to bunch. A partial prohibition produces the same simplification because no agents are in the strict regime. Nonmonetary sanctions differ because they impose real resource costs. Even when a fixed nonmonetary sanction is chosen optimally, those costs create an additional force toward a higher threshold, all else equal, so that fewer agents incur the sanction. Finally, mandates introduce an offsetting consideration: because the mandated technology reduces external harm, a more effective mandate makes the strict regime more socially valuable and therefore pushes toward a lower threshold relative to an otherwise identical nonmonetary sanction. Table 1 summarizes these results.

6 Illustrations from Private Law

This section illustrates our results by applying them to two canonical line-drawing settings in private law: the due-care standard under negligence law and the standard for determining a breach of a contract. In both cases, we determine the optimal line (due care or contract breach) and the optimal damages attached to crossing it.

6.1 Due Care Thresholds and Negligence Liability

Under a negligence regime, an actor is liable for damages for any accident she causes if she fails to take due care. Implementing a negligence regime therefore requires

Table 1: Upper-edge effects and optimal thresholds across applications

Legal setting	Upper-edge effect $W_S(\tilde{x}) - W_B(\tilde{x}; t)$	Implication for the optimal threshold
Optimal fixed monetary sanction	0	The optimal threshold satisfies $\phi'(t) = v_x^B$ and depends only on the welfare effects for continuing bunchers.
Partial prohibition	None; $\tilde{x} = 1$	There is no upper-edge switching effect, so the optimal threshold again satisfies $\phi'(t) = v_x^B$.
Optimal fixed nonmonetary sanction	$\phi(t) - \phi(\tilde{x})$	The real resource cost of the sanction creates an additional force toward a higher threshold.
Fixed-cost mandate	$\phi(t) - \phi(\tilde{x}) + \Delta_\phi(\tilde{x})$	The mandate's harm-reduction effect makes remaining in the strict regime more socially valuable, creating a force toward a lower threshold.

specifying both a due-care standard and damages rule. While in some models of negligence the due-care standard is determined based on particular facts for a given individual, in other cases the due-care standard is based on reasonable conduct for similarly situated actors. In the latter set of cases, the law must draw a generally applicable line to determine whether the actor is deemed negligent. The analysis below applies our general results to this setting: the damages schedule governs both behavior among actors who remain negligent and the stakes associated with choosing between satisfying the due-care standard and remaining negligent.

6.1.1 Setup

Actors choose how much care to exercise. Care is denoted by $x \in [0, 1]$, with higher values of x corresponding to less care. An accident occurs with probability $\pi(x)$, where $\pi(\cdot)$ is differentiable, weakly increasing, and weakly convex. Each accident generates external harm $h > 0$.

The law specifies a due-care threshold t . An actor who chooses $x \leq t$ is non-negligent (the lax regime), while an actor who chooses $x > t$ is negligent (the strict regime). If an accident occurs, an actor who is negligent pays damages $d(x) > 0$, whereas an actor who is non-negligent pays no damages. Because an accident occurs with probability $\pi(x)$, the expected private burden generated by negligence liability is expected liability:

$$c(x) = \pi(x)d(x).$$

Suppose, in addition, that each accident involving a negligent actor generates litigation cost $\lambda > 0$, corresponding to the cost of determining the proper damages award.¹¹ We treat damages as transfers from defendants to victims and therefore do not count them as a direct social cost. For ease of exposition, we assume risk neutrality by both individual actors and society.

At an action x in the lax regime, expected private payoff and expected social welfare are respectively

$$v(x; x^*) \quad \text{and} \quad v(x; x^*) - h\pi(x),$$

At an action x in the strict regime, expected private payoff and expected social welfare are respectively

$$v(x; x^*) - \pi(x)d(x) \quad \text{and} \quad v(x; x^*) - \pi(x)(h + \lambda).$$

6.1.2 Optimal Negligence Damages

Consider first behavior conditional on remaining in the strict regime. An actor with $x^* > t$ who chooses not to bunch at the due-care threshold solves

$$x_S(x^*) \in \arg \max_{x > t} \{v(x; x^*) - \pi(x)d(x)\},$$

¹¹Administrative costs associated with determining whether conduct was negligent can take a different form if they arise even when no accident occurs. We consider an extension that incorporates such costs in the Appendix.

with interior first-order condition

$$v_x(x_S(x^*); x^*) = \pi'(x_S(x^*))d(x_S(x^*)) + \pi(x_S(x^*))d'(x_S(x^*)). \quad (18)$$

The right-hand side is the marginal effect of x on expected liability. Conditional on acting negligently, expected social welfare is $v(x; x^*) - \pi(x)(h + \lambda)$, so the socially optimal strict-regime action satisfies

$$v_x(x; x^*) = (h + \lambda)\pi'(x).$$

Accordingly, the damages rule implements the socially optimal strict-regime action when

$$\frac{d}{dx} [\pi(x)d(x)] = (h + \lambda)\pi'(x).$$

Integrating both sides, for some constant a ,

$$\pi(x)d(x) = (h + \lambda)\pi(x) + a. \quad (19)$$

Equivalently,

$$d(x) = h + \lambda + \frac{a}{\pi(x)}$$

for $\pi(x) > 0$.

Changing a shifts expected liability without changing its slope, so it does not affect behavior conditional on remaining negligent. Instead, it governs sorting at the due-care threshold by changing the stakes associated with bunching at t rather than remaining negligent. In the notation of Proposition 2, $(h + \lambda)\pi(x)$ plays the role of $c_0(x)$ and a plays the role of the action-independent monetary fine or subsidy. Thus, once the slope of expected liability is chosen according to (19), the optimal value of a follows directly from our general stakes result.

For an actor with $x^* > t$, social welfare from bunching is

$$W_B(x^*; t) = v(t; x^*) - h\pi(t),$$

while social welfare from remaining negligent is

$$W_S(x^*) = v(x_S(x^*); x^*) - (h + \lambda)\pi(x_S(x^*)).$$

Under (19), the corresponding private payoff from remaining negligent is $W_S(x^*) - a$, whereas bunching yields $v(t; x^*)$. Proposition 2 therefore implies that the optimal a makes the marginal type socially indifferent between bunching and remaining negligent:

$$W_B(\tilde{x}; t) = W_S(\tilde{x}).$$

The marginal type is also privately indifferent,

$$v(t; \tilde{x}) = W_S(\tilde{x}) - a.$$

Combining these two conditions with the expression for W_B gives

$$a^*(t) = -h\pi(t). \tag{20}$$

Substituting (20) into (19) gives optimal expected liability:

$$\pi(x)d^*(x; t) = h[\pi(x) - \pi(t)] + \lambda\pi(x). \tag{21}$$

Expected damages should therefore equal the sum of (1) the incremental expected harm from choosing x rather than complying with the due-care threshold and (2) the expected litigation cost. The actor is not charged for the baseline expected harm $h\pi(t)$ that would have remained had the actor complied with the due-care standard.¹²

This result is closely related to prior work showing how the causation requirement can limit a negligent actor's liability to the incremental accident risk created by falling short of due care (Grady, 1983; Kahan, 1989). In expected terms, the actor is charged

¹²Although $a^*(t)$ is negative whenever $\pi(t) > 0$, the damages award remains nonnegative. Because π is weakly increasing, $d^*(x; t) = \lambda + h\left(1 - \frac{\pi(t)}{\pi(x)}\right) \geq \lambda > 0$ for every $x > t$ with $\pi(x) > 0$. The strict regime is therefore never privately more attractive than the lax regime at a common action. In particular, no type with $x^* < t$ will move above the threshold to enter the strict regime, so sorting remains one-sided, from above.

not for the full accident loss associated with action x , but for the difference between expected accident losses at x and at the due-care threshold t , namely $h[\pi(x) - \pi(t)]$. Thus, the rule implements a causal approach to negligence damage awards: the actor is responsible only for the portion of expected harm attributable to the failure to comply with the legal standard.

Dividing (21) by $\pi(x)$, damages conditional on an accident occurring are

$$\begin{aligned} d^*(x; t) &= \lambda + h \left(1 - \frac{\pi(t)}{\pi(x)} \right) \\ &= \lambda + h \frac{\pi(x) - \pi(t)}{\pi(x)}. \end{aligned} \tag{22}$$

Optimal damages are thus equal to harm, h , multiplied by the share of accident risk attributable to the actor's noncompliance, plus litigation costs. In particular, the ratio $[\pi(x) - \pi(t)]/\pi(x)$ can be interpreted as the probability that the accident is attributable to the actor's noncompliance.¹³

6.1.3 Optimal Due-Care Threshold

Let $\tilde{x}(t) \equiv \tilde{x}(t, a^*(t))$ denote the marginal type under the optimal damages rule. Because $a^*(t)$ is chosen according to Proposition 2, the marginal type is socially indifferent between bunching and remaining negligent. The upper-edge effect is therefore zero, and Corollary 1 applies directly. Here,

$$\frac{\partial W_B(x^*; t)}{\partial t} = v_x(t; x^*) - h\pi'(t),$$

so the Corollary implies

$$\int_t^{\tilde{x}(t)} [v_x(t; x^*) - h\pi'(t)] dF(x^*) = 0.$$

¹³If courts could identify the accidents that compliance would have prevented, they could impose full damages only in those cases and no damages otherwise. When courts observe only that an accident occurred and the actor's level of care, imposing this fraction of full harm in every accident case produces the same expected liability; see Shavell (1985).

Equivalently, any interior optimal due-care threshold satisfies

$$v_x^B = h\pi'(t), \quad (23)$$

where, as in (16), v_x^B denotes the average marginal cost among buncchers of further deviating from their privately optimal level of care. Thus, once damages are chosen optimally, the due-care threshold has the same form as the optimal threshold in the fixed-monetary-sanction application, with expected accident harm $h\pi(x)$ taking the place of the general externality $\phi(x)$.

Condition (23) is a line-drawing analogue of the Hand Rule. In the canonical economic analysis of negligence, there is a single relevant socially optimal level of care, and setting the due-care standard at that level can induce the actor to comply and thereby implement efficient precaution (Brown, 1973; Shavell, 1980b). In particular, the individually differentiated standard equates each actor's private burden of additional precaution with the expected accident loss that the precaution would avert. Our setting departs from that benchmark because actors differ in the private burden of care, while the law must apply a common due-care standard. There is therefore no single threshold that reproduces the individualized first-best level of care for every actor.¹⁴ Condition (23) gives the corresponding second-best analogue of the usual Hand calculation. The marginal expected accident loss from relaxing the standard, $h\pi'(t)$, plays the role of the expected loss avoided by additional precaution, while $v_x(t; x^*)$ measures the marginal private burden of requiring an actor of type x^* to exercise that precaution. Because the same line applies to heterogeneous actors, however, the relevant burden is the average burden among the actors whose behavior the line actually changes—that is, the buncchers. The optimal standard therefore performs the familiar Hand balancing on average over precisely the subpopulation induced by the due-care standard to alter its conduct.

¹⁴Prior work on heterogeneous injurers emphasizes the distortions created by a common negligence standard and shows how richer, graduated liability schedules can induce actors with different care costs to self-select into their individually optimal levels of care (Miceli, 2006); see also Emons and Sobel (1991) on the design of liability rules for nonidentical actors. We instead retain the institutional constraint that negligence law draws a single due-care line and ask where that line should be placed when the damages schedule is chosen jointly with it.

Finally, note that litigation costs do not enter this threshold condition directly because they arise only when an accident occurs. They enter instead through the damages award, via the λ term in (22). By changing the attractiveness of remaining negligent, litigation costs alter the size of the bunching region and thus the composition of the actors whose precaution costs enter the average in (23).

6.2 Contractual Performance Thresholds and Expectation Damages

Contract remedies provide a deterministic counterpart to the negligence application. A contract must specify what counts as conforming performance even when the performing party's burden of satisfying that standard is privately observed and cannot be incorporated into a fully contingent agreement.¹⁵ The resulting line-drawing problem is to choose jointly a common performance floor and the damages schedule that applies above it. The application illustrates how the general results derived above map onto familiar contract-theory logic: the slope of damages governs performance conditional on breach, while the level of damages governs sorting between conforming performance and breach.

6.2.1 Setup

A performing party chooses an action $x \in [0, 1]$, with higher values representing poorer performance along a dimension such as quality, quantity, timeliness, or completeness. The contract law default rule specifies a performance threshold t : performance $x \leq t$ is conforming, while performance $x > t$ constitutes breach. The party's privately preferred action absent contract remedies for breach is indexed by x^* . We

¹⁵This may occur because the parties use a standardized contract, because the relevant contingency is difficult to describe or verify ex ante, or because the contract leaves gaps to be filled by trade usage or legal default rules. In such settings, damages can substitute for more completely specified contingent terms by shaping conduct when the promised performance is not supplied. This is the familiar role of contract remedies emphasized by Shavell (1980a), Shavell (1984), and Cooter and Eisenberg (1985). Related work studies quality and material-breach thresholds as devices for shaping investment, rejection, and termination incentives; see Stremitzler (2012), Ganglmair (2017), and Ganglmair (2019).

assume that the burden of better performance is privately observed and cannot be used to tailor the contractual threshold to each party's type.

Let $\phi(x)$ denote the loss that performance x imposes on the counterparty and that the performing party does not internalize absent a contract remedy. We assume that $\phi(\cdot)$ is differentiable, weakly increasing, and weakly convex. If breach triggers damages $d(x)$, the private burden generated by the damages rule is $d(x)$, so private payoff above the line is $v(x; x^*) - d(x)$. We treat damages as a transfer between the parties, but suppose that each enforced breach generates a fixed net litigation or enforcement cost $\lambda \geq 0$. The social welfare contribution from a type x^* is therefore $v(x; x^*) - \phi(x)$ for conforming performance and $v(x; x^*) - \phi(x) - \lambda$ for breach.¹⁶

6.2.2 Optimal Damages

A party of type $x^* > t$ that chooses to breach chooses

$$x_S(x^*) \in \arg \max_{x > t} \{v(x; x^*) - d(x)\},$$

with interior first-order condition

$$v_x(x_S(x^*); x^*) = d'(x_S(x^*)). \quad (24)$$

Because the enforcement cost λ is fixed once breach occurs, the socially optimal action conditional on breach satisfies

$$v_x(x; x^*) = \phi'(x).$$

The optimal damages rule therefore implements efficient performance conditional on breach when

$$d'(x) = \phi'(x).$$

¹⁶The analysis focuses on ex post performance and abstracts from any effect of the remedial rule on earlier reliance or relationship-specific investment. Contract remedies may also shape those ex ante incentives, and expectation damages need not implement first-best investment in every environment. See, for example, Rogerson (1984), Edlin and Reichelstein (1996), and Stremitzler (2012).

Hence, any optimal damages rule must satisfy

$$d(x) = \phi(x) + a, \quad (25)$$

where a is a constant. As in negligence, the slope governs behavior conditional on breach, while the constant governs the stakes associated with conforming rather than breaching. The difference is that contract enforcement costs are incurred whenever breach occurs and therefore affect only the constant, rather than the slope, of the optimal payment schedule.

Social welfare from performing at the threshold and from remaining in breach are, respectively,

$$W_B(x^*; t) = v(t; x^*) - \phi(t)$$

and

$$W_S(x^*) = v(x_S(x^*); x^*) - \phi(x_S(x^*)) - \lambda.$$

Because a is a transfer that affects the choice between conforming and breaching, but not performance conditional on breach, Proposition 2 implies that its optimal value makes the marginal type socially indifferent between conforming and breaching. This, along with the fact that the marginal type is also privately indifferent between the two, implies

$$a^*(t) = \lambda - \phi(t). \quad (26)$$

The optimal breach-contingent payment is therefore

$$d^*(x; t) = \phi(x) - \phi(t) + \lambda. \quad (27)$$

The term $\phi(x) - \phi(t)$ has the familiar expectation-damages interpretation. The threshold t defines the contractual performance entitlement, and $\phi(x) - \phi(t)$ is the additional loss caused by performance x rather than performance at that threshold. Thus, absent enforcement costs, applying Proposition 2 recovers the standard expectation-damages logic: damages make the performing party internalize the counterparty's loss from deficient performance (Shavell, 1980a, 1984; Cooter and Eisen-

berg, 1985). When $\lambda > 0$, the payment also includes the social cost generated by enforcement.

6.2.3 Optimal Performance Threshold

Let $\tilde{x}(t) \equiv \tilde{x}(t, a^*(t))$ denote the marginal type under the optimal payment schedule. Because $a^*(t)$ eliminates the upper-edge effect, Corollary 1 implies that any interior optimal threshold satisfies

$$\int_t^{\tilde{x}(t)} [v_x(t; x^*) - \phi'(t)] dF(x^*) = 0. \quad (28)$$

Equivalently,

$$\phi'(t) = v_x^B. \quad (29)$$

Condition (29) determines how demanding the contractual performance obligation should be. Tightening the standard gives the counterparty better performance from parties who are induced to satisfy it, but requires those parties to bear additional performance costs. The optimal performance threshold equates the marginal counterparty loss averted by better performance, $\phi'(t)$, with the average marginal burden of compliance among the parties induced by the threat of damages to satisfy the contractual standard.

7 Conclusion

Our results show that line-drawing problems across different areas of law share a common logic. Some actors change their behavior just enough to remain on the favorable side of the line; they bunch at the line. Moving the line affects these actors by changing both the burden of adjusting their behavior and the harm that their adjusted behavior causes. It also affects the marginal actor at the upper edge of the bunching group, who may switch between bunching and remaining in the strict regime. The optimal line balances the effects on those who continue to bunch against what society gains or loses when that marginal actor switches between these

two outcomes.

We also show that a monetary fine or subsidy, imposed in addition to any other legal consequence, can be used to adjust the private stakes associated with crossing the line and thereby shape how many actors bunch at the line. Because such a payment changes who bunches without changing the behavior of those who remain in the strict regime, it can be set so that the marginal actor produces the same social welfare whether that actor bunches or remains in the strict regime. The upper-edge effect is then zero. Therefore, when the government can impose such a payment, the optimal line depends only on the effects of moving it on actors who continue to bunch. As a result, although the same threshold formally applies to an entire population, the welfare calculus that determines its optimal placement can be concentrated on a much narrower group: the actors whose behavior the line actually changes. Moreover, that group is itself endogenous to legal design. The consequences attached to crossing the line affect which actors choose to bunch and therefore whose costs and benefits enter the optimal line-drawing calculation.

In contrast, when the government cannot freely optimize the private stakes associated with crossing the line, the upper-edge effect remains relevant, and differences among legal regimes enter the optimal line-drawing problem through a common consideration: the social value of leaving the marginal actor in the strict regime rather than inducing that actor to bunch. A nonmonetary sanction makes remaining in the strict regime more costly because it consumes real resources; a mandate may make it more valuable because it reduces external harm; and a partial prohibition eliminates this switching margin altogether. The upper-edge effect thus provides a common way to understand why otherwise similar lines should be drawn differently depending on the legal consequences attached to crossing them.

We illustrate this common structure through two familiar private-law doctrines. In negligence law, we show that the optimal due-care standard is a population version of the Hand Rule: it balances the marginal reduction in accident harm against the average burden of precaution among the actors whose conduct the due-care standard induces to change. Contract law provides a deterministic counterpart, with the optimal performance threshold balancing the value of better performance against

the average burden among the parties induced to comply. The optimal damages in both settings also share a common structure. In both cases, the line defines the relevant baseline and the damages are set to reflect the additional harm caused by conduct beyond that baseline rather than the total harm associated with the actor’s conduct.

Our analysis relies on a relatively stylized setting and can be extended in a number of directions. For example, we assume that the law draws a line. We do not compare line drawing to other approaches, such as sliding scales (e.g., ?) or uncertain standards (e.g., Craswell and Calfee (1986)). In addition, we assume that the law draws a single line rather than using multiple categories. We are not aware of work that analyzes the optimal number of categories within a legal system. We also assume that the law draws the line with respect to the externality-causing behavior but often this behavior is not observable and the law must use imperfect proxies, which may introduce additional considerations (e.g., Jacobsen et al., 2020; Ito and Sallee, 2018). Finally, different potential lines sometimes differ meaningfully with respect to how difficult it is to classify actors near the threshold; in the appendix, we show that our baseline results extend in a straightforward manner in the presence of such classification costs.

Declaration of generative AI use.

Generative AI was used to aid in the analysis and writing of this work. The authors take full responsibility for the contents of the published article.

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Appendix

Classification Costs

The baseline analysis abstracts from administrative costs associated with determining which side of a line an agent's conduct falls. Such costs are likely to be greatest for close cases. Conduct far from the threshold may be easy to classify, whereas conduct very near the line may require more precise measurement, factfinding, or adjudication. This consideration is especially relevant in our setting because the line may itself induce a positive mass of agents to locate exactly at the threshold.

To capture this possibility, let $k(r, t) \geq 0$ denote the social cost of classifying an agent whose realized action lies a distance $r \geq 0$ from a threshold located at t . We assume that k is continuously differentiable and weakly decreasing in distance,

$$k_r(r, t) \leq 0. \quad (30)$$

Thus, classification is weakly more costly as conduct approaches the line. The second argument allows the difficulty of classification to depend on where the line itself is drawn: some thresholds may be easier to observe or administer than others even holding fixed an agent's distance from the line. We assume that classification costs do not enter agents' private payoffs and therefore do not affect sorting directly.

For an agent whose privately preferred action lies below the threshold, realized conduct is x^* and its distance from the line is $t - x^*$. A bunching agent selects $x = t$ and therefore lies exactly at the line. An agent who is in the strict regime selects $x_S(x^*)$ and lies a distance $x_S(x^*) - t$ above it. Aggregate welfare is therefore

$$\begin{aligned} W(t) = & \int_0^t [W_L(x^*) - k(t - x^*, t)] dF(x^*) \\ & + \int_t^{\tilde{x}(t)} [W_B(x^*; t) - k(0, t)] dF(x^*) \\ & + \int_{\tilde{x}(t)}^1 [W_S(x^*) - k(x_S(x^*) - t, t)] dF(x^*). \end{aligned} \quad (31)$$

To simplify the threshold condition, define the administrative welfare effect of

moving the line while holding agents' behavioral categories fixed as

$$\begin{aligned}
A(t, \tilde{x}) := & - \int_0^t [k_r(t - x^*, t) + k_t(t - x^*, t)] dF(x^*) \\
& - \int_t^{\tilde{x}} k_t(0, t) dF(x^*) \\
& + \int_{\tilde{x}}^1 [k_r(x_S(x^*) - t, t) - k_t(x_S(x^*) - t, t)] dF(x^*). \tag{32}
\end{aligned}$$

The first line captures the effect of moving the threshold relative to agents whose privately preferred actions already fall in the lax regime. The second captures any change in the cost of classifying agents who continue to bunch. The third captures the corresponding effect for agents who remain in the strict regime.

Differentiating (31) with respect to t gives

$$\begin{aligned}
\frac{\partial W}{\partial t} = & \int_t^{\tilde{x}} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) + A(t, \tilde{x}) \\
& + \frac{\partial \tilde{x}}{\partial t} [W_B(\tilde{x}; t) - k(0, t) - W_S(\tilde{x}) + k(x_S(\tilde{x}) - t, t)] f(\tilde{x}). \tag{33}
\end{aligned}$$

Accordingly, any interior optimal threshold satisfies

$$\begin{aligned}
& \int_t^{\tilde{x}} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) + A(t, \tilde{x}) \\
& = \frac{\partial \tilde{x}}{\partial t} [W_S(\tilde{x}) - W_B(\tilde{x}; t) + k(0, t) - k(x_S(\tilde{x}) - t, t)] f(\tilde{x}). \tag{34}
\end{aligned}$$

Equation (34) preserves the decomposition in Proposition 1. The first term is the welfare effect from changing the behavior of inframarginal bunchers. The term $A(t, \tilde{x})$ captures the administrative effect of moving the line relative to agents who remain in their existing behavioral categories. The right-hand side is the upper-edge effect, now measured after accounting for the difference in classification costs between bunching and remaining in the strict regime.

Classification costs also modify the optimal private stakes for crossing the line. Suppose the government can choose the fixed monetary fine or subsidy τ introduced

in Section 4. Because τ remains a pure transfer and does not affect behavior conditional on being in the strict regime, it affects welfare only through the marginal type $\tilde{x}(t, \tau)$. Differentiating welfare with respect to τ gives

$$\begin{aligned} \frac{\partial W(t, \tau)}{\partial \tau} = & [W_B(\tilde{x}; t) - k(0, t) - W_S(\tilde{x}) + k(x_S(\tilde{x}) - t, t)] \\ & \times f(\tilde{x}) \frac{\partial \tilde{x}}{\partial \tau}. \end{aligned} \quad (35)$$

Thus, any interior optimal choice of τ satisfies

$$W_B(\tilde{x}; t) - k(0, t) = W_S(\tilde{x}) - k(x_S(\tilde{x}) - t, t). \quad (36)$$

As in Proposition 2, the monetary payment is chosen so that society is indifferent between the marginal type bunching and remaining in the strict regime. The difference is that this comparison now incorporates the administrative costs generated by the two outcomes. Because bunching places the agent exactly at the line whereas being in the strict regime generally places the agent some distance above it, (30) implies

$$k(0, t) \geq k(x_S(\tilde{x}) - t, t).$$

Classification costs therefore make bunching less socially attractive at the margin. Because increasing τ expands the set of agents who bunch, this consideration creates a force toward lower private stakes for crossing the line and hence toward fewer agents locating at the threshold.

Combining (36) with the threshold condition yields an analogue to Corollary 1. Because the optimally chosen τ eliminates the upper-edge effect after accounting for classification costs, any interior optimal threshold satisfies

$$\int_t^{\tilde{x}} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) + A(t, \tilde{x}) = 0. \quad (37)$$

Thus, the basic division of labor between the two policy instruments continues to apply. The stakes for crossing the line can be used to make the marginal sorting

decision socially optimal, including the administrative cost of creating an additional close case. The location of the line is then determined by its effects on continuing bunchers together with the administrative consequences of placing the threshold at one location rather than another.

A useful benchmark is the case in which different locations for the threshold are not intrinsically easier or harder to administer. Suppose classification costs depend only on distance from the line, so that $k(r, t) = k(r)$ and therefore $k_t(r, t) = 0$. In that case, (32) simplifies to

$$A(t, \tilde{x}) = - \int_0^t k'(t - x^*) dF(x^*) + \int_{\tilde{x}}^1 k'(x_S(x^*) - t) dF(x^*). \quad (38)$$

Because $k'(\cdot) \leq 0$, the first term is weakly positive: raising the threshold moves the line farther away from the conduct of agents who remain in the lax regime and thereby makes those cases easier to classify. The second term is weakly negative: raising the threshold moves the line closer to agents who remain in the strict regime and thereby makes those cases harder to classify. Which effect dominates depends on the distribution of conduct near the threshold on the two sides of the line and on how rapidly classification costs decline with distance. In the limiting case in which classification costs are positive only for agents whose conduct lies exactly at the line, the administrative cost is generated entirely by the bunching mass. If the per-agent cost at the line is k , equation (37) further simplifies to

$$\int_t^{\tilde{x}} \frac{\partial W_B(x^*; t)}{\partial t} dF(x^*) + kf(t) = 0. \quad (39)$$

The additional term reflects the classification-cost saving associated with agents at the lower edge of the bunching interval becoming naturally located in the lax regime as the threshold is raised. Changes in the bunching mass at the upper edge do not appear separately because, with the monetary stakes chosen optimally, the upper-edge welfare effect is zero after accounting for classification costs.

In the more general case in which some threshold locations are intrinsically easier to administer than others, the k_t terms in (32) introduce an additional consideration.

For example, one line may correspond to an observable or easily verified distinction whereas another may require costly fact-finding even for conduct the same distance away. The optimal threshold then reflects not only the welfare effects from changing agents' behavior but also these differences in the administrative difficulty of alternative lines.